

108 AI — Desktop Agent

On-premise desktop agent that runs on the client's machine and provides OS-level capabilities (file system, shell, code editing, git, clipboard, desktop automation) to AI agents running on the 108 AI platform.

How It Works

The desktop agent connects to the 108 AI Gateway via WebSocket and waits for action requests from AI assistants. When an action is requested, it executes it locally (within configured security boundaries) and returns the result.

```
AI Assistant --> 108 AI Gateway --> WebSocket --> Desktop Agent --> OS
                                     |
                               File System / Shell / Git / Desktop
```

Installation

Prerequisites

- Node.js 20+
- npm 10+

From Source

```
cd apps/local-agent
npm install
npm run build
```

Docker (alternative)

```
docker build -t 108ai-desktop .
docker run -d \
  --name 108ai \
  -e GATEWAY_URL=wss://api.108ai.dev \
  -e TENANT_ID=your-tenant-uuid \
  -e AUTH_TOKEN=your-auth-token \
  -v /path/to/data:/data \
  108ai-desktop
```

Configuration

On first run, an interactive setup wizard will guide you. Configuration is stored at:

```
~/.108ai/config.json
```

Config Options

Option	Description	Default
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gatewayUrl	Gateway WebSocket URL	(required)
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authToken	Authentication token	(required)
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tenantId	Tenant UUID	(required)
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- allowedDirectories | Directories the agent can access | Documents, Desktop, Downloads
- autoStart | Start on system boot | false
- maxActionsPerMinute | Rate limit | 10
- desktopEnabled | Enable desktop automation | false
- shellEnabled | Enable shell/process actions | false
- gitEnabled | Enable git actions | true
- gitAllowPush | Allow git push without explicit approval | false
- gitAllowDestructive | Allow destructive git (reset) without explicit approval | false

CLI Arguments

```
108ai --gateway-url wss://api.108ai.dev --tenant-id <UUID> --token <token>
```

CLI arguments override config file values.

Capabilities

Filesystem

Action	Description	Risk
filesystem.readFile	Read a text file (max 10MB)	read_only
filesystem.writeFile	Write content to a file	low_risk
filesystem.listDirectory	List files in a directory	read_only
filesystem.searchFiles	Search files by glob pattern	read_only
filesystem.watchDirectory	Watch for file changes	read_only
filesystem.getFileInfo	Get file metadata	read_only

Shell

Action	Description	Risk
shell.execute	Execute command, return stdout/stderr/exit	high_risk
shell.executeStream	Long-running command with streaming output	high_risk
shell.terminate	Kill a running process	low_risk
shell.getRunning	List active agent-spawned processes	read_only

Code

Action	Description	Risk
code.edit	Diff-based edit (old_string -> new_string)	low_risk
code.editMulti	Multiple atomic edits in one file	low_risk
code.write	Write entire file (new files or full rewrite)	low_risk
code.readRange	Read line range with line numbers	read_only

Git

Action	Description	Risk
git.status	Working tree status	read_only
git.diff	Staged + unstaged diff	read_only
git.log	Recent commits	read_only
git.commit	Stage + commit	low_risk
git.branch	Create/switch/list branches	low_risk
git.stash	Stash/pop changes	low_risk

git.blame | Line-level blame | read_only
git.push | Push to remote (gated) | high_risk
git.reset | Reset (soft/mixed only, gated) | high_risk

Search

Action	Description	Risk
search.grep	Regex search with context (ripgrep-like)	read_only
search.glob	File pattern matching	read_only
search.find	Recursive find by name/type	read_only

Web

Action	Description	Risk
web.fetch	GET URL, return content	low_risk
web.search	Web search via API	read_only

Process

Action	Description	Risk
process.start	Start long-running command (gated by shellEnabled + approval)	high_risk
process.stop	Stop a running process	low_risk
process.list	List running processes	read_only
process.logs	Tail process logs	read_only

MCP (via Extensions)

MCP is currently exposed via the extensions layer (CLI + local UI), not as mcp.* WebSocket actions.

- CLI: /mcp list|add|install|start|tools|test
- Config: ~/.108ai/mcp.yml

Clipboard

Action	Description	Risk
clipboard.read	Read clipboard content	read_only
clipboard.write	Write to clipboard	low_risk

System

Action	Description	Risk
system.openUrl	Open URL in default browser	low_risk
system.openFile	Open file with default app	low_risk
system.showNotification	Show OS notification	read_only
system.getSystemInfo	Get OS/memory/disk info	read_only

Desktop (opt-in)

Action	Description	Risk
desktop.listWindows	List visible windows	read_only
desktop.readWindow	Read window text via accessibility	read_only
desktop.readFocused	Read focused window content	read_only
desktop.screenshot	Capture window/screen screenshot	read_only

desktop.analyzeScreen | Screenshot + LLM vision analysis | read_only
desktop.getUITree | Get accessibility UI tree | read_only
desktop.focusWindow | Bring window to foreground | low_risk
desktop.scrollWindow | Scroll within a window | low_risk
desktop.typeText | Type text into focused input | high_risk
desktop.clickElement | Click UI element by name | high_risk
desktop.pressHotkey | Press keyboard hotkey | high_risk
desktop.mouseClick | Click at screen coordinates | high_risk

Security

Sandboxing

- File operations are restricted to explicitly configured allowedDirectories
- Path traversal (..) is prevented by resolving to absolute paths
- System directories (Windows, Program Files, /usr, /etc) are always blocked
- Binary files cannot be read
- Shell commands require explicit approval from the gateway

Risk Levels

Level	Behavior
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read-only	Auto-approved (no state change)
low-risk	Auto-approved (minor, reversible state change)
high-risk	Requires gateway approval flag (_approved: true)

Rate Limiting

- Maximum 10 actions per minute (configurable)
- Prevents runaway AI agents from overwhelming the system

Audit Log

All actions are logged to ~/.108ai/audit.log in JSONL format:

```
{"timestamp":"2026-06-09T10:30:00.000Z","action":"filesystem.readFile","params":{"path":"/home/user/doc.txt"},"result":"..."}
```

Authentication

- WebSocket connection requires a valid auth token
- Token is validated by the gateway on connection
- Heartbeat every 30s to detect disconnection

Auto-Start

To add the agent to system startup:

```
# The installer module handles platform-specific registration:  
# - Windows: Registry Run key  
# - macOS: LaunchAgent (dev.108ai.desktop)  
# - Linux: systemd user service (108ai-desktop.service)
```

Development

```
# Run in development mode (with hot reload)
npm run dev

# Build
npm run build

# Run built version
npm start
```

Interactive shell & extensions

The REPL shell (pnpm dev) supports custom commands, skills, persona agents, and MCP servers under ~/.108ai/.

Feature CLI	
Terminal dashboard	/ui dashboard
Command palette	/palette [query]
Web dashboard	/ui web → http://127.0.0.1:7891
Persona agents	/agent use <name>
Daily triage	/trriage, /morning

See docs/USER-GUIDE.md and docs/SECURITY-RUNBOOK.md.

AI Model Routing

The 108 AI platform routes requests through LiteLLM with the following tiers:

Tier	Primary	Fallback	Use Case
fast-cheap	DeepSeek V3	Qwen3-8B	Chat, simple tasks, high volume
balanced	DeepSeek R1	Qwen3-32B	Complex reasoning, multi-step
powerful	Qwen3-235B	DeepSeek R1	Architecture, critical analysis
coding	DeepSeek V3	Qwen3-30B	Code generation, review
vision	Qwen-VL-Max	DeepSeek V3	Screenshot analysis, OCR
embedding	Alibaba text-embedding-v3	OpenAI ada-3-small	RAG, semantic search

All model calls are tracked for per-tenant billing.